Muhammad Ali Waris Khan

AI / RAG Engineer — Generative AI Developer

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Professional Summary

AI and Retrieval-Augmented Generation (RAG) developer with hands-on experience building and deploying production-oriented LLM applications using Gemini AI, LangChain, FAISS, and Streamlit. Skilled in semantic search, vector databases, conversational AI, prompt engineering, neural reranking, and multi-stage retrieval pipelines. Developed end-to-end AI systems including document intelligence platforms, YouTube transcript assistants, academic research copilots, and ATS resume analyzers with real-world deployment and modular software architecture.

Technical Skills

Programming: Python
AI / LLMs: Gemini AI, LangChain, Prompt Engineering, Conversational AI, RAG Pipelines
Retrieval & Vector Search: FAISS, Semantic Search, Dense Embeddings, MMR Retrieval, Cross-Encoder Reranking
Machine Learning / NLP: Sentence Transformers, Hugging Face Embeddings, NLP Preprocessing
Frontend & Deployment: Streamlit, Streamlit Cloud
Databases & Processing: Vector Databases, PDF Parsing, Transcript Processing
Libraries & Tools: PyMuPDF, yt-dlp, youtube-transcript-api, sentence-transformers, dotenv
Software Engineering: Modular Architecture, API Integration, Exception Handling, Stateful Session Management

Projects

1. Context-Aware RAG Chatbot with Gemini

(Python, LangChain, Gemini AI, FAISS, Streamlit, PyMuPDF)

- Built and deployed a production-style conversational RAG chatbot enabling interaction with multiple PDF documents using Gemini AI and LangChain
- Implemented semantic retrieval pipeline using Gemini embeddings and FAISS vector database for context-aware document search
- Engineered conversational memory system with multi-turn dialogue support using ConversationalRetrievalChain and ConversationBufferMemory
- Optimized retrieval quality using MMR-based retrieval strategy to improve contextual diversity and reduce redundant retrievals
- Developed explainable AI response system with page-level citations and source traceability for grounded answer generation
- Designed modular and scalable AI architecture with reusable ingestion, chunking, embedding, vector storage, and retrieval components

2. AI Resume Analyzer with ATS Scoring

(Python, Gemini AI, Streamlit, PyMuPDF, Regex NLP)

- Developed an AI-powered ATS resume analyzer that evaluates resumes against job descriptions using NLP-based keyword analysis
- Built automated PDF resume parsing and preprocessing pipeline with robust validation and exception handling
- Engineered custom ATS scoring algorithm using keyword extraction, skill matching, and missing-skill detection workflows
- Integrated Gemini AI to generate contextual resume optimization feedback and recruiter-style improvement recommendations
- Implemented explainable scoring interface displaying matched skills, missing keywords, and compatibility insights
- Designed modular HR-tech application architecture with secure API handling and production-oriented UI workflows

3. Advanced YouTube RAG Chatbot with Transcript Intelligence

(Python, Gemini AI, LangChain, FAISS, Sentence Transformers, Streamlit)

- Built an advanced conversational YouTube RAG assistant enabling semantic interaction with video transcripts using Gemini AI
- Implemented transcript intelligence pipeline with automated metadata extraction, timestamp-aware chunking, and semantic indexing
- Engineered multi-stage retrieval architecture combining FAISS semantic search, MMR retrieval, and Cross-Encoder reranking

- Integrated conversational memory and streaming LLM responses for real-time context-aware multi-turn interaction
- Developed explainable timestamp-based citation system linking generated answers directly to original video segments
- Implemented persistent vector database storage with incremental transcript indexing for scalable retrieval workflows

4. AI Research Paper Assistant with Literature Review Generation

(Python, Gemini AI, LangChain, FAISS, Hugging Face, Streamlit)

- Developed an AI-powered academic research assistant supporting semantic search, literature review generation, and conversational paper analysis
- Built multi-document RAG pipeline using Hugging Face embeddings and FAISS vector databases for semantic academic retrieval
- Implemented multi-stage retrieval architecture with Cross-Encoder reranking for improved retrieval precision and grounded responses
- Engineered automated research summarization, methodology extraction, and literature synthesis workflows using Gemini AI
- Designed hallucination-aware prompt engineering strategies enforcing context-grounded academic question answering
- Built modular production-oriented AI architecture with explainable citation retrieval and robust API validation workflows

Education

Masters in Artificial Intelligence
College of Electrical and Mechanical Engineering, NUST, Islamabad

Links

- github.com/maw-khan
- linkedin.com/in/muhammad-ali-waris-khan-97a8ba146
- https://mawk-portfolio.vercel.app/